01204325

การสื่อสารข้อมูลและเครือข่ายคอมพิวเตอร์

(Data Communication and Computer Networks)

Information Theory

ผศ.ดร. ประวิทย์ ชุมชู

Email:sfengwpc@src.ku.ac.th

Facebook:https://www.facebook.com/sfengpwc

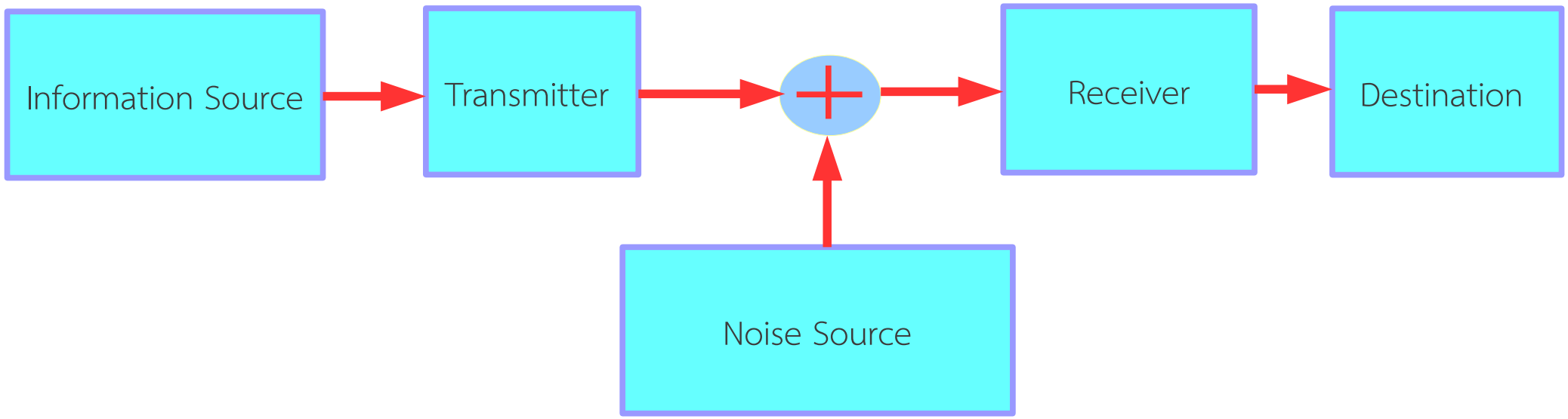
Mobile Number:0865343850

สอน W13:16-10304, F13-16-17106

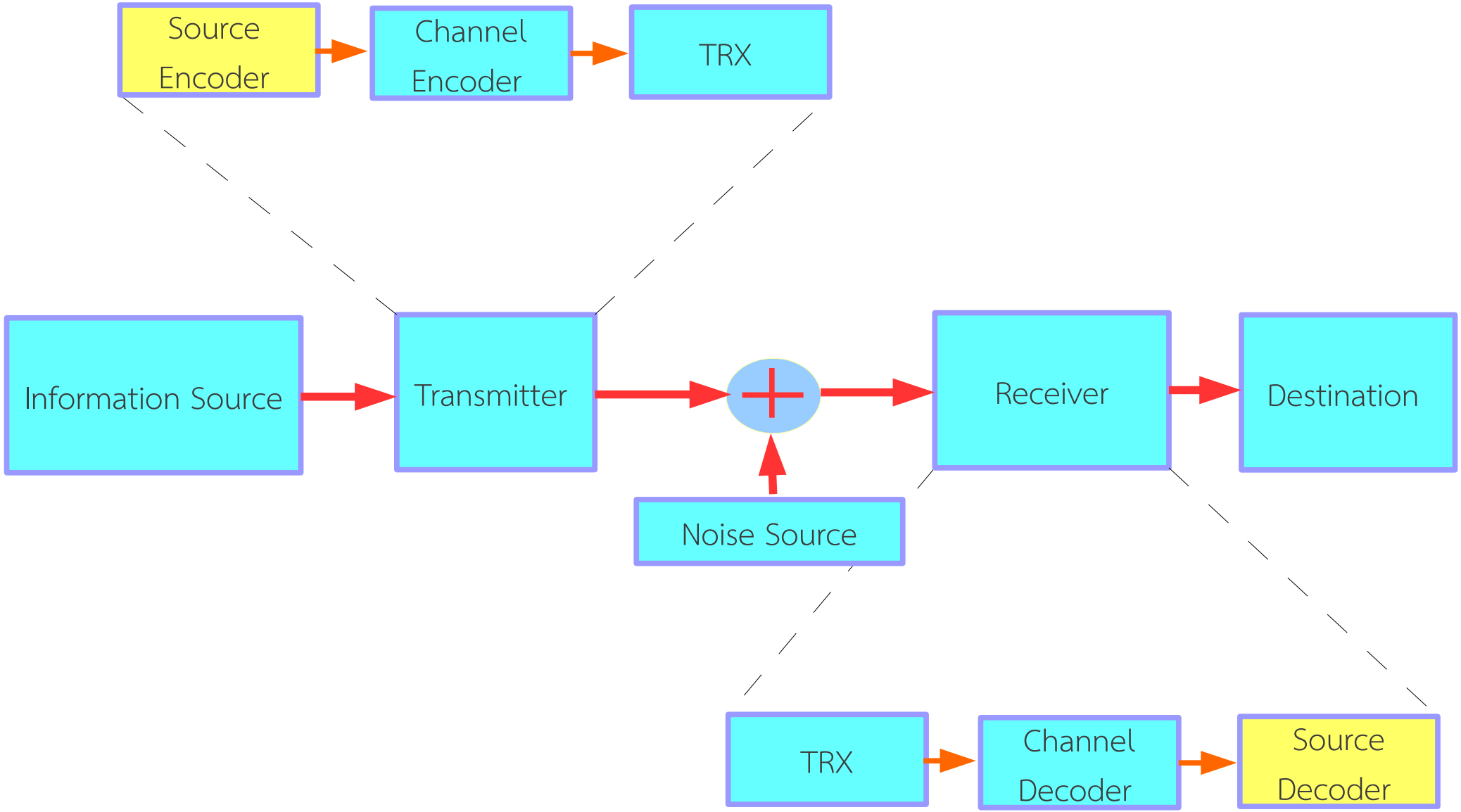
Outline

- ตัวอย่างการรับส่งข้อมูล
- Information Source (แหล่งข้อมูล)
- Information Measurement (การวัดข้อมูล)
- Entropy(เอนโทรปี)
- Source Coding (การเข้ารหัสข้อมูล)
- Data Compression(การบีบอัดข้อมูล)
- สรุป

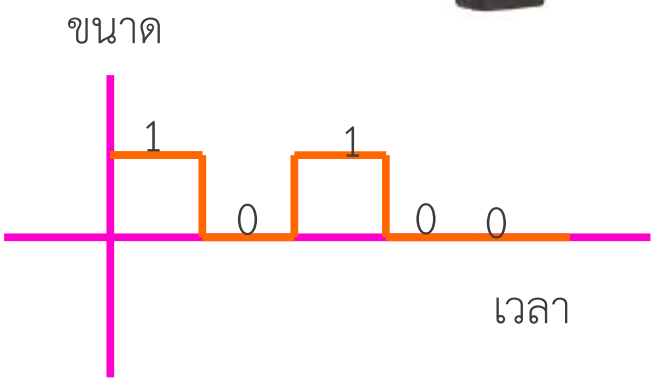
การสื่อสารข้อมูลพื้นฐาน

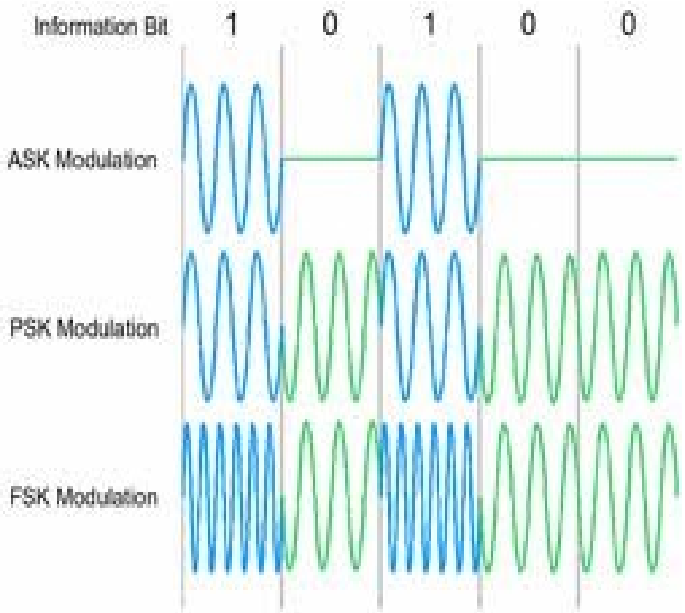


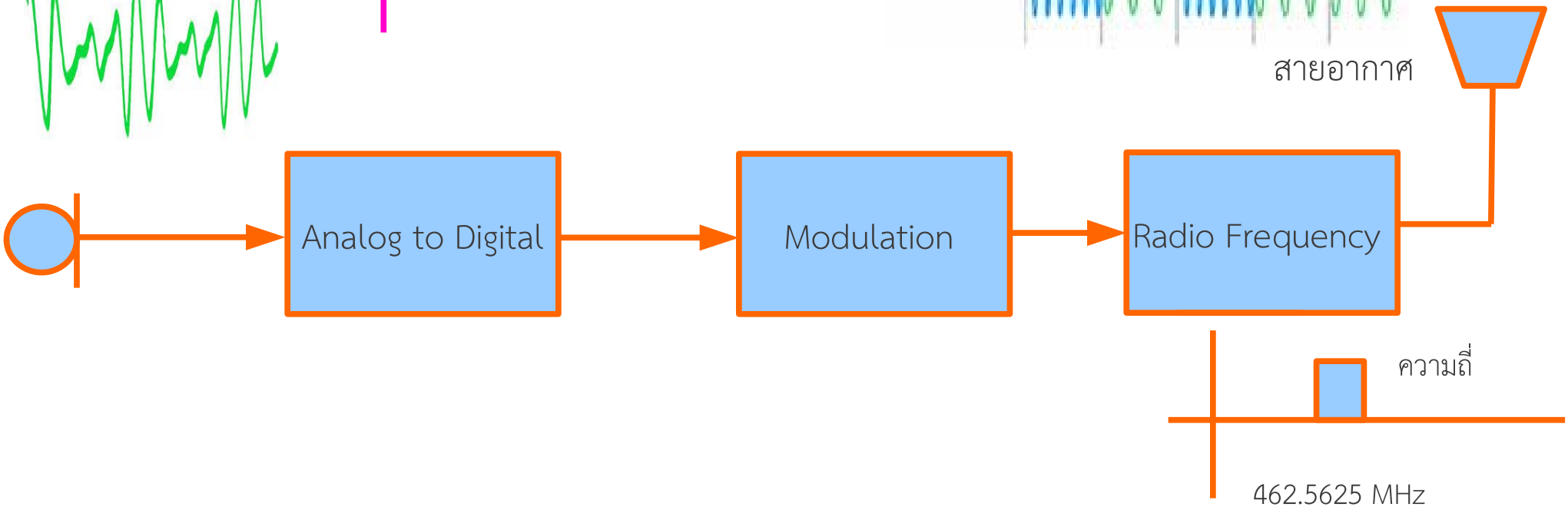
การสื่อสารข้อมูลพื้นฐาน



ตัวอย่างการสื่อสาร Digital



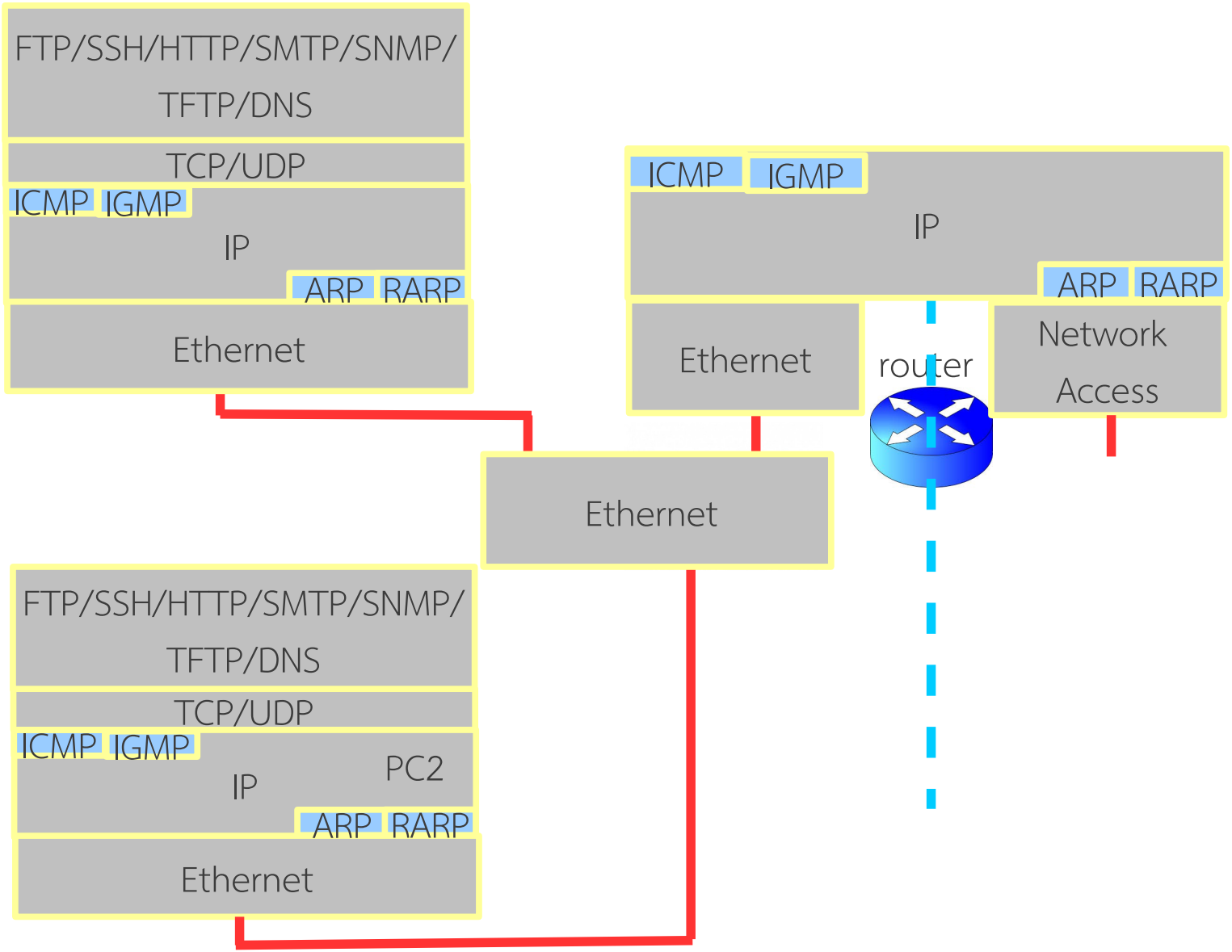




จากตัวอย่างจงตอบคำถามต่อไปนี้

- Information Source= ?
- Transmitter= ?
- Receiver =?
- Receiver = ?
- Destination = ?

ตัวอย่างเครือข่าย Ethernet



จากตัวอย่างจงตอบคำถามต่อไปนี้

- Information Source= ?
- Transmitter= ?
- Receiver =?
- Receiver = ?
- Destination = ?

1	139022531	192.168.43.2	192.168.43.1	Source port: 36198 Destination port: 7890
2	139022531		D-Link_0a:4f:3d (RA)	Acknowledgement, Flags=.....

```
▶ Frame 1: 206 bytes on wire (1648 bits), 206 bytes captured (1648 bits)
▶ Prism capture header
▶ IEEE 802.11 Data, Flags: .......T
▶ Logical-Link Control
▶ Internet Protocol Version 4, Src: 192.168.43.2 (192.168.43.2), Dst: 192.168.43.1 (192.168.43.1)
▶ User Datagram Protocol, Src Port: 36198 (36198), Dst Port: 7890 (7890)
▼ Data (2 bytes)
    Data: 4869
```

```
0000  44 00 00 00 90 00 00 00  61 74 68 31 00 00 00 00   D....... ath1....
0010  00 00 00 00 00 00 00 00  44 00 01 00 00 00 04 00   ........ D.......
0020  15 b5 1c 00 44 00 02 00  00 00 04 00 00 a8 c5 1d   ....D... ........
0030  44 00 03 00 00 00 04 00  06 00 00 00 44 00 04 00   D....... ....D...
0040  00 00 04 00 4b 00 00 00  00 00 00 00 00 00 00 00   ....K... ........
0050  00 00 00 00 44 00 06 00  00 00 04 00 f1 ff ff ff   ....D... ........
0060  44 00 07 00 00 00 04 00  a6 ff ff ff 44 00 08 00   D....... ....D...
0070  00 00 04 00 6c 00 00 00  44 00 09 00 00 00 04 00   ....l... D.......
0080  01 00 00 00 44 00 0a 00  00 00 04 00 3e 00 00 00   ....D... ....>...
0090  08 01 2c 00 f0 e7 7e d3  2f 14 00 17 9a 0a 4f 3d   ..,...~. /.....O=
00a0  f2 e7 7e d3 2f 14 80 11  aa aa 03 00 00 00 08 00   ..~./... ........
00b0  45 00 00 1e b1 d8 40 00  40 11 b1 a2 c0 a8 2b 02   E.....@. @.....+.
00c0  c0 a8 2b 01 8d 66 1e d2  00 0a 33 e4 48 69         ..+..f.. ..3.Hi
```

ข้อมูลของ Source ?

```
▶ Frame 4: 461 bytes on wire (3688 bits), 461 bytes captured (3688 bits)
▶ Ethernet II, Src: Vmware_76:af:f4 (00:0c:29:76:af:f4), Dst: Vmware_be:60:70 (00:0c:29:be:60:70)
▶ Internet Protocol Version 4, Src: 172.16.197.4 (172.16.197.4), Dst: 172.16.197.3 (172.16.197.3)
▶ Transmission Control Protocol, Src Port: 49163 (49163), Dst Port: http (80), Seq: 1, Ack: 1, Len: 407
▼ Hypertext Transfer Protocol
  ▼ GET / HTTP/1.1\r\n
    ▶ [Expert Info (Chat/Sequence): GET / HTTP/1.1\r\n]
      Request Method: GET
      Request URI: /
      Request Version: HTTP/1.1
    Accept: image/jpeg, application/x-ms-application, image/gif, application/xaml+xml, image/pjpeg, applicatio
    Accept-Language: en-us\r\n
    User-Agent: Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 6.1; Trident/4.0; SLCC2; .NET CLR 2.0.50727; .NE
    Accept-Encoding: gzip, deflate\r\n
    Host: 172.16.197.3\r\n
    Connection: Keep-Alive\r\n
    \r\n
    [Full request URI: http://172.16.197.3/]
```

```
0000  00 0c 29 be 60 70 00 0c  29 76 af f4 08 00 45 00   ..).`p.. )v....E.
0010  01 bf 01 c3 40 00 80 06  15 4d ac 10 c5 04 ac 10   ....@... .M......
0020  c5 03 c0 0b 00 50 fe 0d  8b a5 ab 08 73 aa 50 18   .....P.. ....s.P.
0030  40 29 6f 93 00 00 47 45  54 20 2f 20 48 54 54 50   @)o...GE T / HTTP
0040  2f 31 2e 31 0d 0a 41 63  63 65 70 74 3a 20 69 6d   /1.1..Ac cept: im
0050  61 67 65 2f 6a 70 65 67  2c 20 61 70 70 6c 69 63   age/jpeg , applic
0060  61 74 69 6f 6e 2f 78 2d  6d 73 2d 61 70 70 6c 69   ation/x- ms-appli
0070  63 61 74 69 6f 6e 2c 20  69 6d 61 67 65 2f 67 69   cation,  image/gi
0080  66 2c 20 61 70 70 6c 69  63 61 74 69 6f 6e 2f 78   f, appli cation/x
0090  61 6d 6c 2b 78 6d 6c 2c  20 69 6d 61 67 65 2f 70   aml+xml,  image/p
```

ข้อมูลของ Source ?

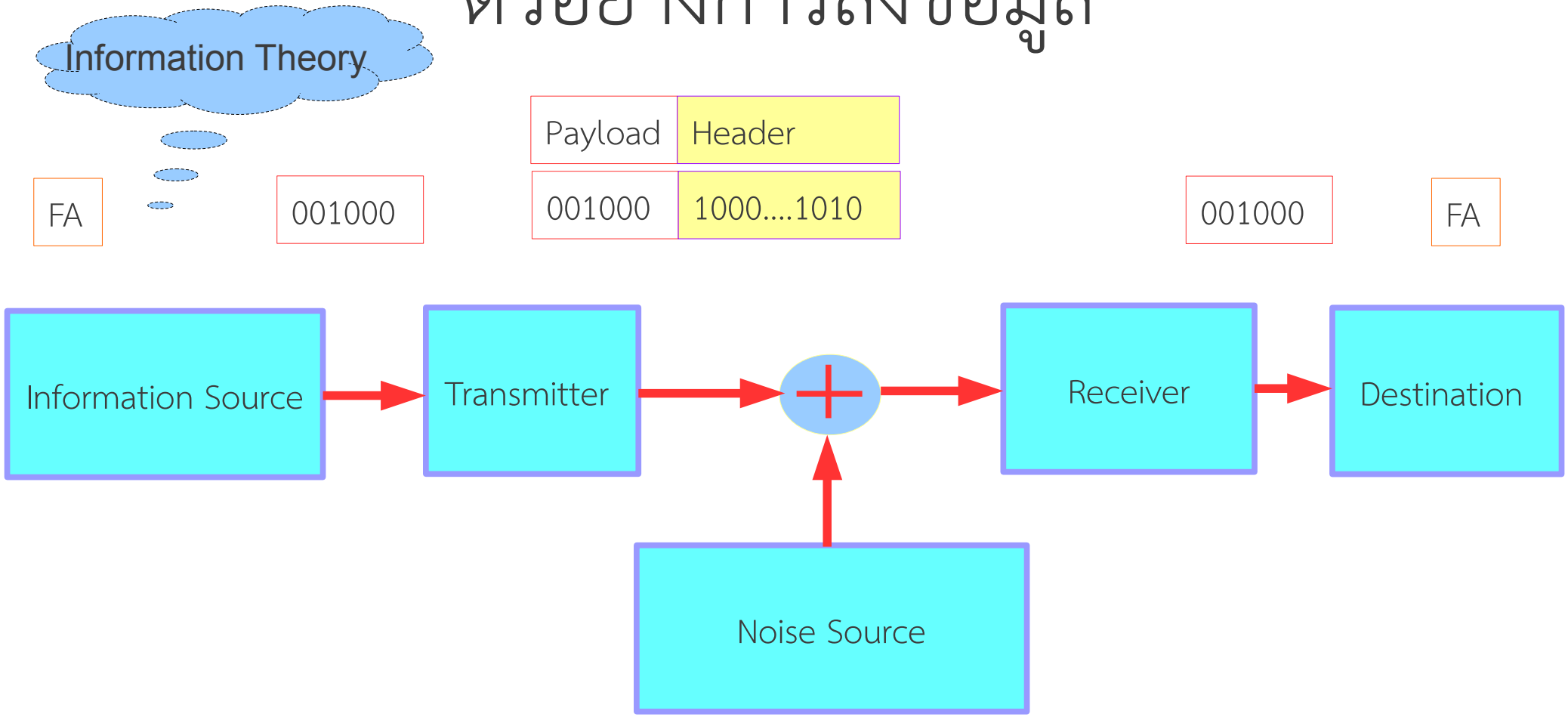
ตัวอย่างการส่งข้อมูล (Data)

- ข้อมูล “ABCDF”

ตัวอย่าง Source Encoder

Source	codeword
A	000
B	001
C	010
D	011
F	100

ตัวอย่างการส่งข้อมูล



ตัวอย่างการส่งข้อมูล

0101..111000111..	011010110010110010110010110010

จงเขียนข้อมูลที่รับได้

ตัวอย่างการส่งข้อมูล (Data)

- ข้อมูลอนาล็อก (Analog)

ตัวอย่าง Source Encoder

Source (Volt)	codeword
0-1	000
1-2	001
2-3	010
3-4	011
4-5	100
5-6	101
6-7	110
7-8	111

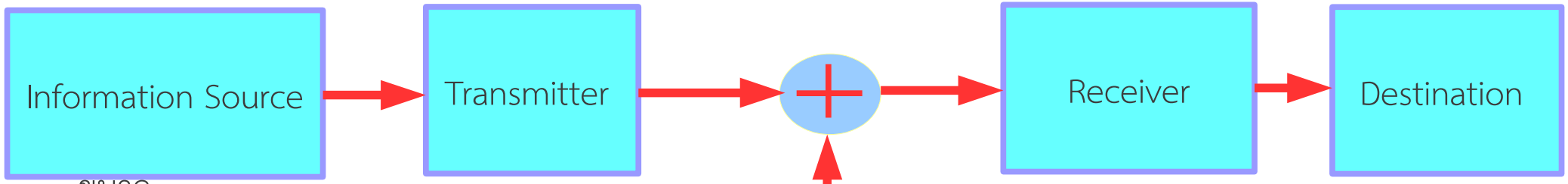
ตัวอย่างการส่งข้อมูล

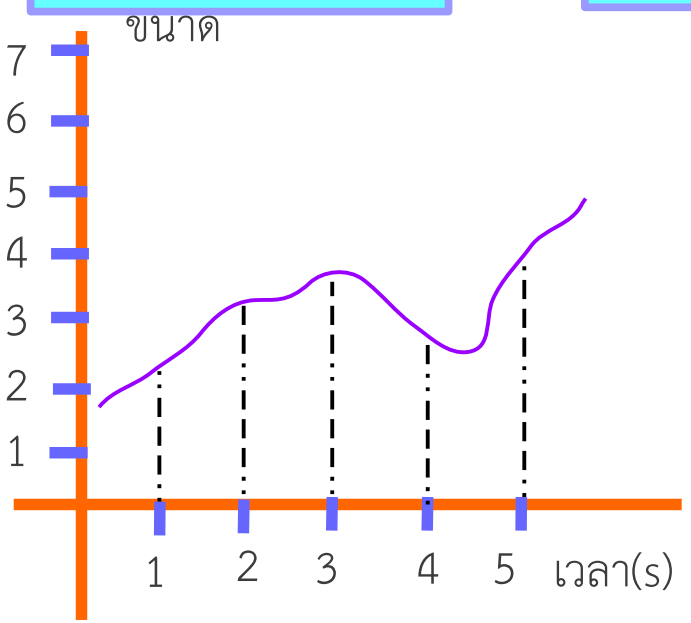
Information Theory

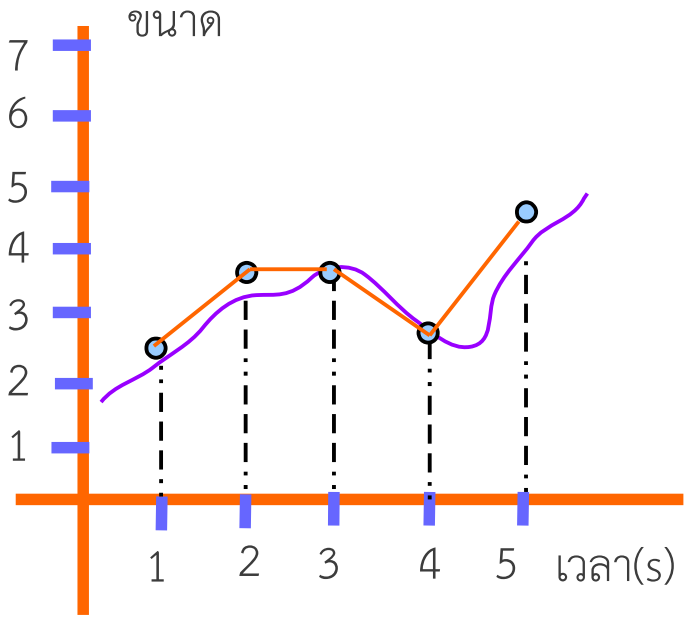
Payload	Header
001010110110010	1....110

001010110110010

001010110110010







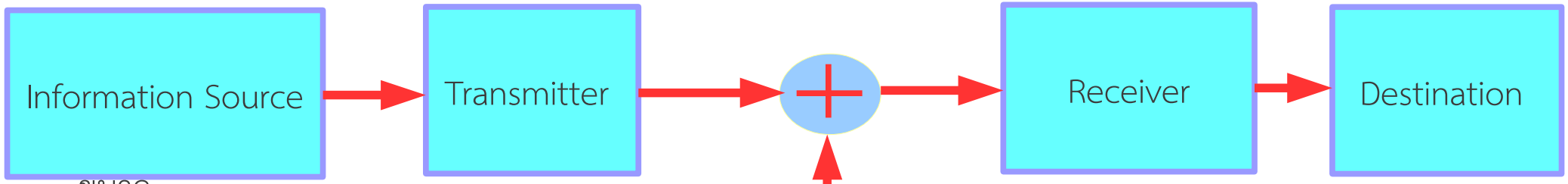
ตัวอย่างการส่งข้อมูล

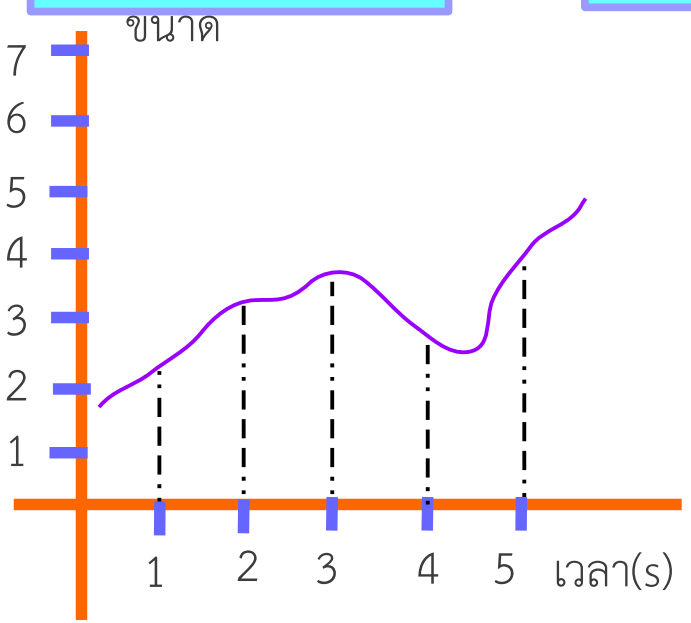
Information Theory

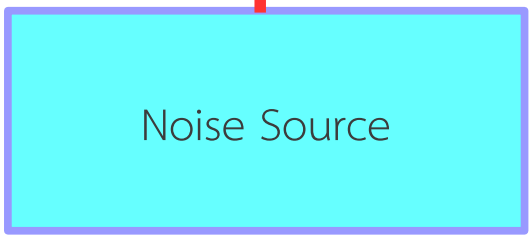
Payload	Header
001010110110010	1....110

001010110110010

001010110110010

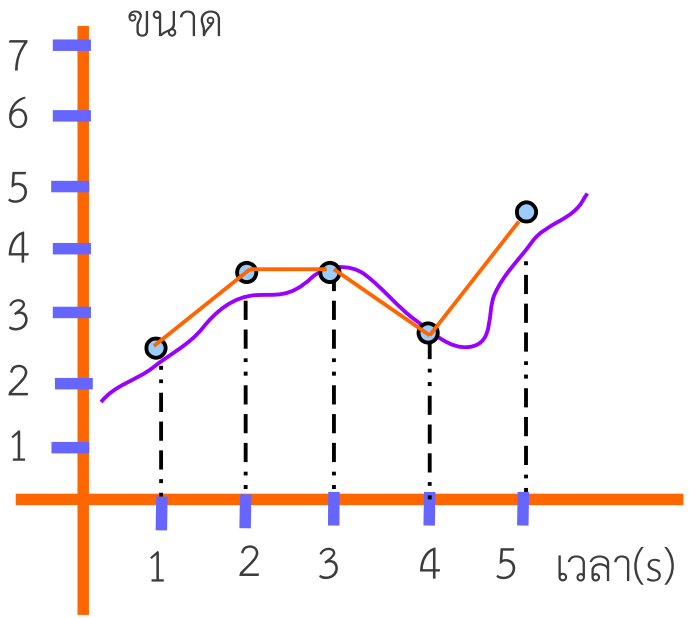






Data Rate=?

Sampling Rate=?

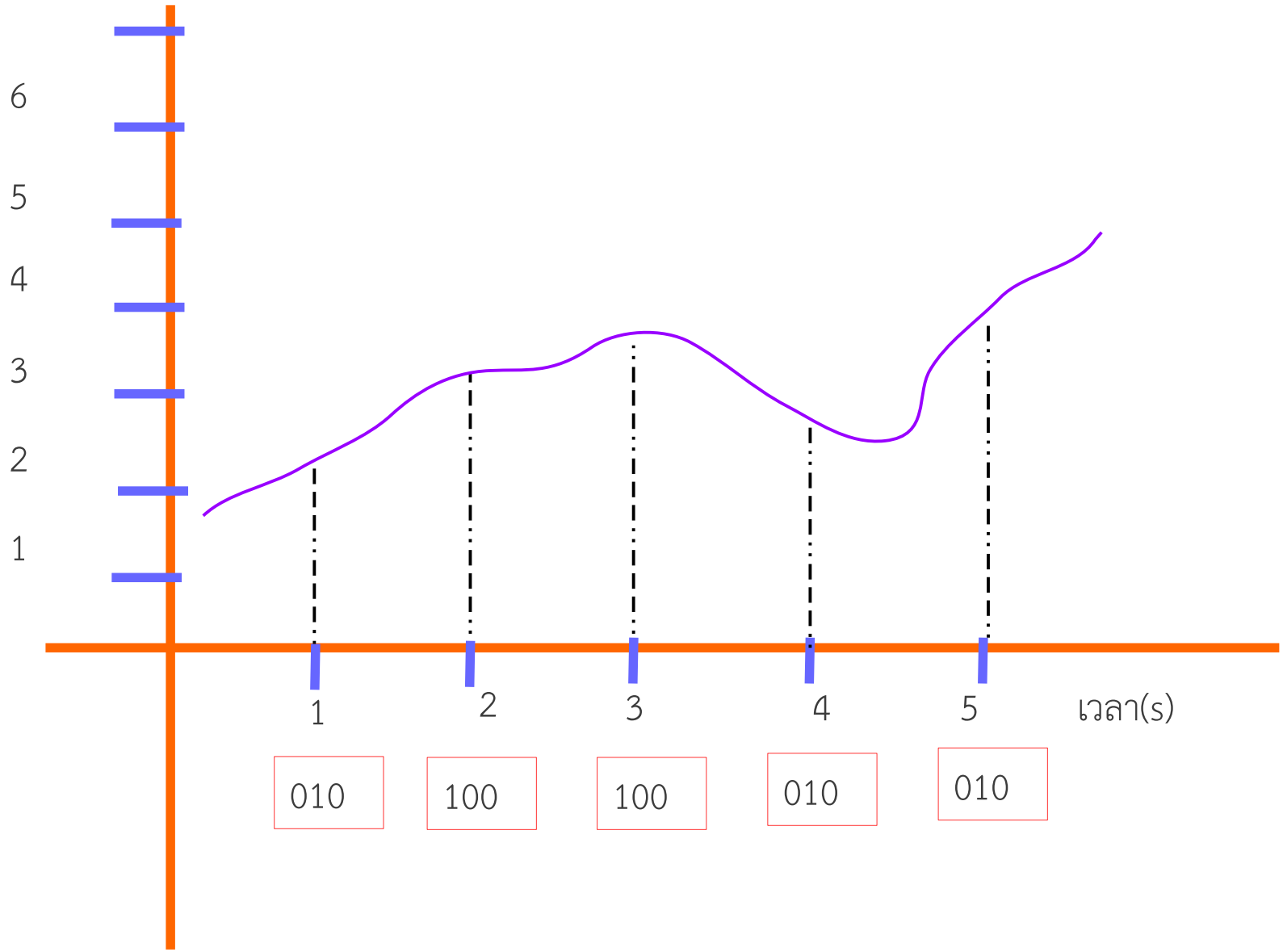


Bit rate และ Sampling Rate

- Bit Rate = จำนวนบิตของข้อมูลที่ส่งต่อวินาที
- Sampling Rate คือ อัตราความถี่ของการสุ่มข้อมูลเช่น
  - 1 Hz/s, Sampling interval = 1/Sampling Rate = 1 วินาที
  - 10 Hz/s, Sampling interval = 1/Sampling Rate =0.1 วินาที

- Nyquist Frequency $> 2f_{max}$
- No Aliasing

Data Rate=?



Data Rate = ?

- ข้อมูล “ABCDF”

Source	codeword
A	000
B	001
C	010
D	011
F	100

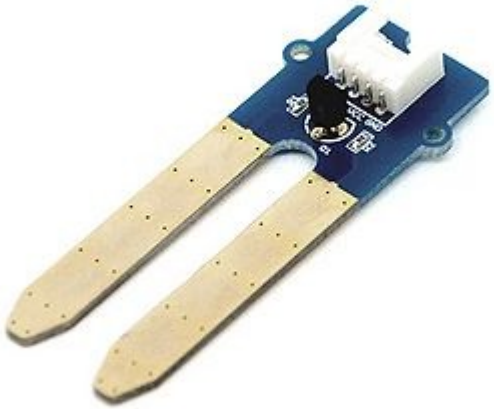
ตัวอย่างการส่งข้อมูล

0101..111000111..	011010110010110010110010110010

จงเขียนข้อมูลที่รับได้

Information Source(1)

- Multimedia
  - Voice
  - Video
  - Sensor
  - Text, Message, Computer Informati
  - Etc.



Information Source (2)

Symbol = x_1, x_2, x_3, x_4

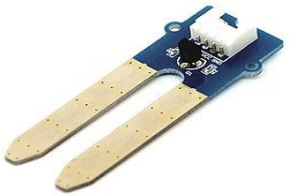
Probability of occur = $P(x_1), P(x_2), P(x_3), P(x_4)$

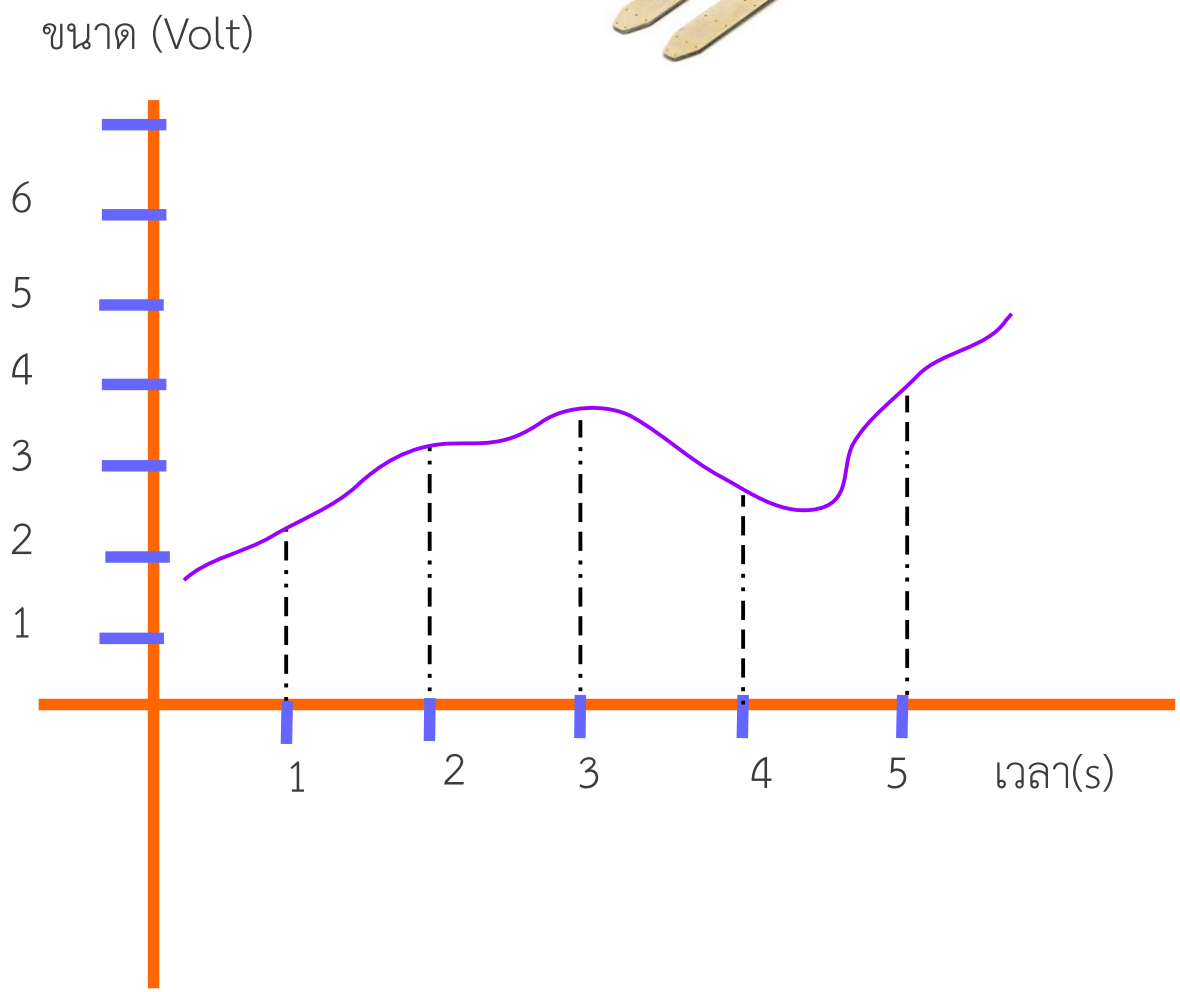
Source(x)	P(x)
A, x_1	1/5
B, x_2	3/6
C, x_3	1/5
D, x_4	0.5/5
F, x_5	0.5/5

Information Source(3)

Source (Volt)	P(x)
0-1, x_1	1/16
1-2, x_2	1/6
2-3, x_3	1/8
3-4, x_4	2/8
4-5, x_5	2/8
5-6, x_6	1/8
6-7, x_7	1/16
7-8, x_8	1/16

Sensors





Information Measurement(1)

- “A measure of the information content of a message can be based on the amount of uncertainty it removes” From: Michael Duck and Richard Read
  - ปริมาณความไม่แน่นอนที่เหตุการณ์นั้นไม่ได้เกิดขึ้น
  - $P(x_1)=0.5$, $P(x_2)=0.5$
  - $P(x_2)=0.8$, $P(x_2)=0.2$
  - $P(x_1)=1$, $P(x_2)=0$

$$I(x_i)=\log_b(1/P(x_i))=-\log_b(P(x_i))$$

มีนตีบ

The term bit can lead to some confusion as it is used on one hand by the communications engineer to represent information content, and on the other hand by the computer scientist to indicate the number of binary digits.

Information Measurement(2)

$$I(x_i) = \log_b(1/P(x_i)) = -\log_b(P(x_i))$$

The choice of the logarithmic base b is equivalent to selecting the unit of information.

When b=2

$$I(x_i) = \log_2(1/P(x_i)) = -\log_2(P(x_i))$$

The standard convention in information theory is for b to be numerically equal to two, in which case the unit of information is called a **bit**.

- $P(x_1)=0.5$, $P(x_2)=0.5$, $I(x_1)=1$, $I(x_2)=1$
- $P(x_2)=0.8$, $P(x_2)=0.2$, $I(x_1)=0.32$, $I(x_2)=2.231$
- $P(x_1)=1$, $P(x_2)=0$, $I(x_1)=0$, $I(x_2)$=Not defined

Information Measurement(3)

Source(x)	P(x)	I(x)
A, x_1	1/5	2.321
B, x_2	3/6	1
C, x_3	1/5	2.321
D, x_4	0.5/5	3.321
F, x_5	0.5/5	3.321

Information Measurement(3)

Source(x)	P(x)	I(x)
A, x_1	1/5	2.321
B, x_2	1/5	2.321
C, x_3	1/5	2.321
D, x_4	1/5	2.321
F, x_5	1/5	2.321

Entropy(1)

- “The average amount of information conveyed by a single symbol within a set of codewords” From: Michael Duck and Richard Read

$$H(x)=\sum_{i=1}^{i=n} P(x_i)\log_2(1/P(x_i)) \qquad bits/symbol$$

Entropy(2)

Source(x)	P(x)
Very Poor, x_1	1/4
Poor, x_2	1/8
Moderate, x_3	1/8
Good, x_4	1/2

$$
\begin{aligned}
H(x) &= \sum_{i=1}^{i=n} P(x_i)\log_2(1/P(x_i)) \\
&= \sum_{i=1}^{i=4} P(x_i)\log_2(1/P(x_i)) \\
&= P(x_1)\log_2(1/P(x_1)) + P(x_2)\log_2(1/P(x_2)) + P(x_3)\log_2(1/P(x_3)) + P(x_4)\log_2(1/P(x_4)) \\
&= 1/4\text{x}\log_2(4) + 1/8\text{x}\log_2(8) + 1/8\text{x}\log_2(8) + 1/2\text{x}\log_2(2) \\
&= 1/2 + 3/8 + 3/8 + 1/2 = 14/8 = 1.75\, bits/symbol
\end{aligned}
$$

From: Michael Duck and Richard Read

Entropy(3)

$$H(x)=\sum_{i=1}^{i=n} P(x_i)\log_2(1/P(x_i)) \qquad bits/symbol$$

Source(x)	P(x)
A, x_1	1/5
B, x_2	3/6
C, x_3	1/5
D, x_4	0.5/5
F, x_5	0.5/5

$$H(x)=?$$

I(x)
2.321
1
2.321
3.321
3.321

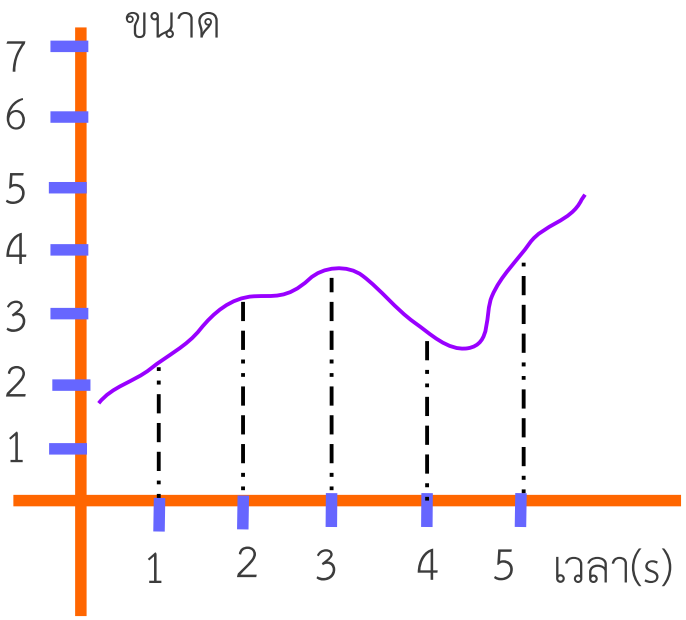
Source Coding(1)

The aim of source coding is to produce a code which, on average, requires the transmission of the maximum amount of information for the fewest binary digits.

A codeword คือ ?

A code คือ ?

Source (Volt)	codeword
0-1	000
1-2	001
2-3	010
3-4	011
4-5	100
5-6	101
6-7	110
7-8	111



Source Coding(2)

The length of a code is the average length of its codewords and is obtained by:

$$L = \sum_{i=1}^{i=n} P(x_i) l_i \qquad bits$$

The efficiency of a code is obtained by dividing the entropy by the average code length:

$$\eta = \frac{H(x)}{L} x100$$

Source Coding(3)

ตัวอย่าง

Source(x)	P(x)	Coseword
Very Poor, x_1	1/4	00
Poor, x_2	1/8	01
Moderate, x_3	1/8	10
Good, x_4	1/2	11

$$L=\sum_{i=1}^{i=n} P(x_i)\, l_i \qquad bits$$

1/4*2+1/8*2+1/8*2+1/2*2 =2

I(x)
2
3
3
1

$$\eta=\frac{H(x)}{L}\, x100$$

H(x)=1/4*2+1/8*3+1/8*3+1/2*1=1.75

Eff=1.75/2*100=87.5 %

Source Coding(4)

ตัวอย่าง

Source(x)	P(x)	Coseword
Very Poor, x_1	1/4	00
Poor, x_2	1/4	01
Moderate, x_3	1/4	10
Good, x_4	1/4	11

$$L=\sum_{i=1}^{i=n} P(x_i)l_i \qquad bits$$

$$\eta=\frac{H(x)}{L}x100$$

Compression(1)

- Data compression seeks to reduce the data transmitted by reducing, or eliminating, any redundancy in the data.
- ตัวอย่างเช่น

Source(x)	P(x)	codeword
A, x_1	1/5	000
B, x_2	3/6	001
C, x_3	1/5	010
D, x_4	0.5/5	011
F, x_5	0.5/5	100

Data rate =?
How to Compress it

Compression(2)

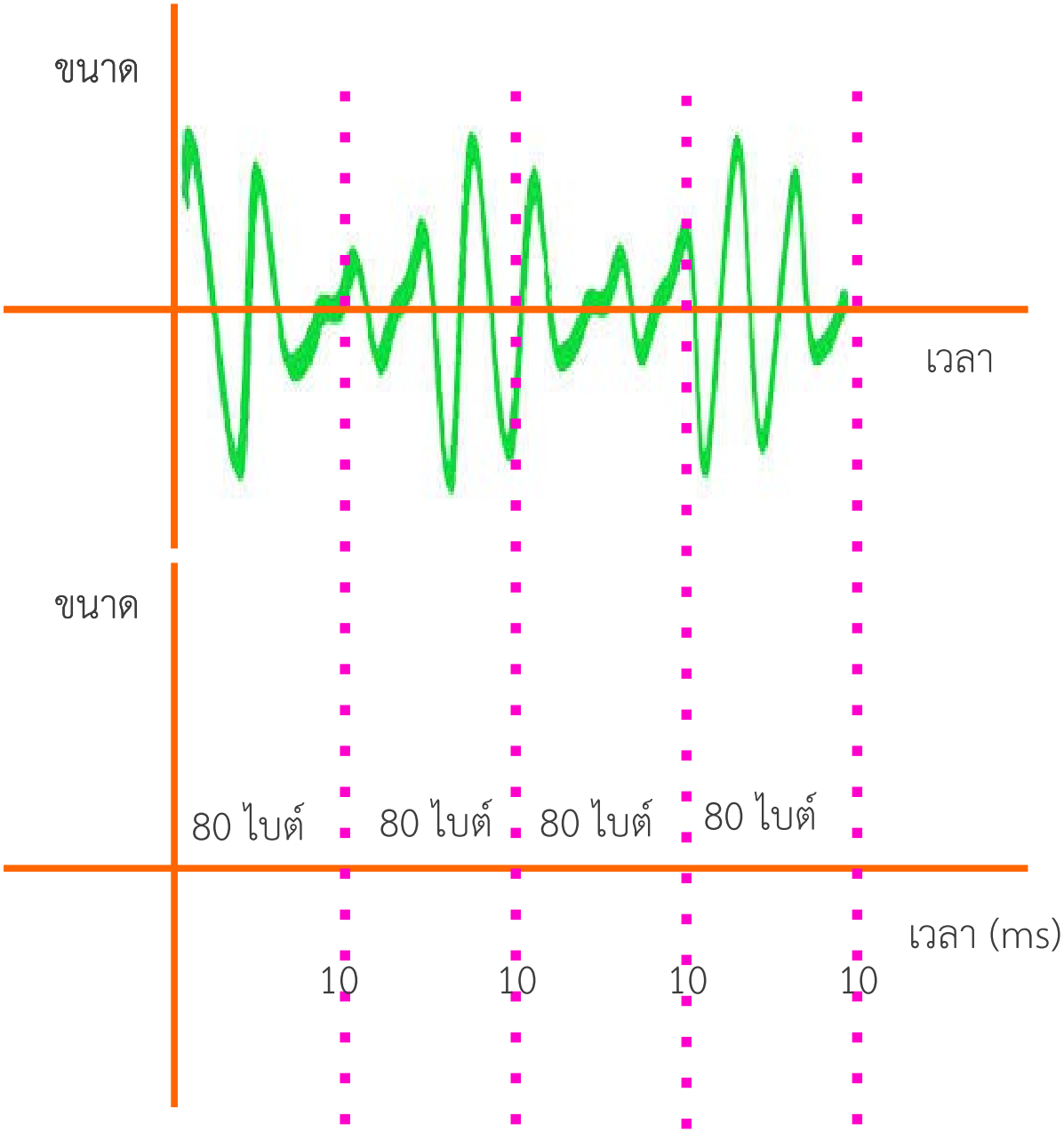
Source(x)	P(x)	codeword	codeword
A, x_1	1/5	000	00
B, x_2	3/6	001	01
C, x_3	1/5	010	10
D, x_4	0.5/5	011	110
F, x_5	0.5/5	100	111

โอกาศเกิดน้อยส่งด้วยจำนวนบิตมากกว่า

AABBCBBCFA = 000 000 001 001 010 001 001 010 100 000 =30 bits
AABBCBBCFA = 00 00 01 01 10 01 01 10 111 01 =21 bits

Next Class: Huffman Code and other complex codes

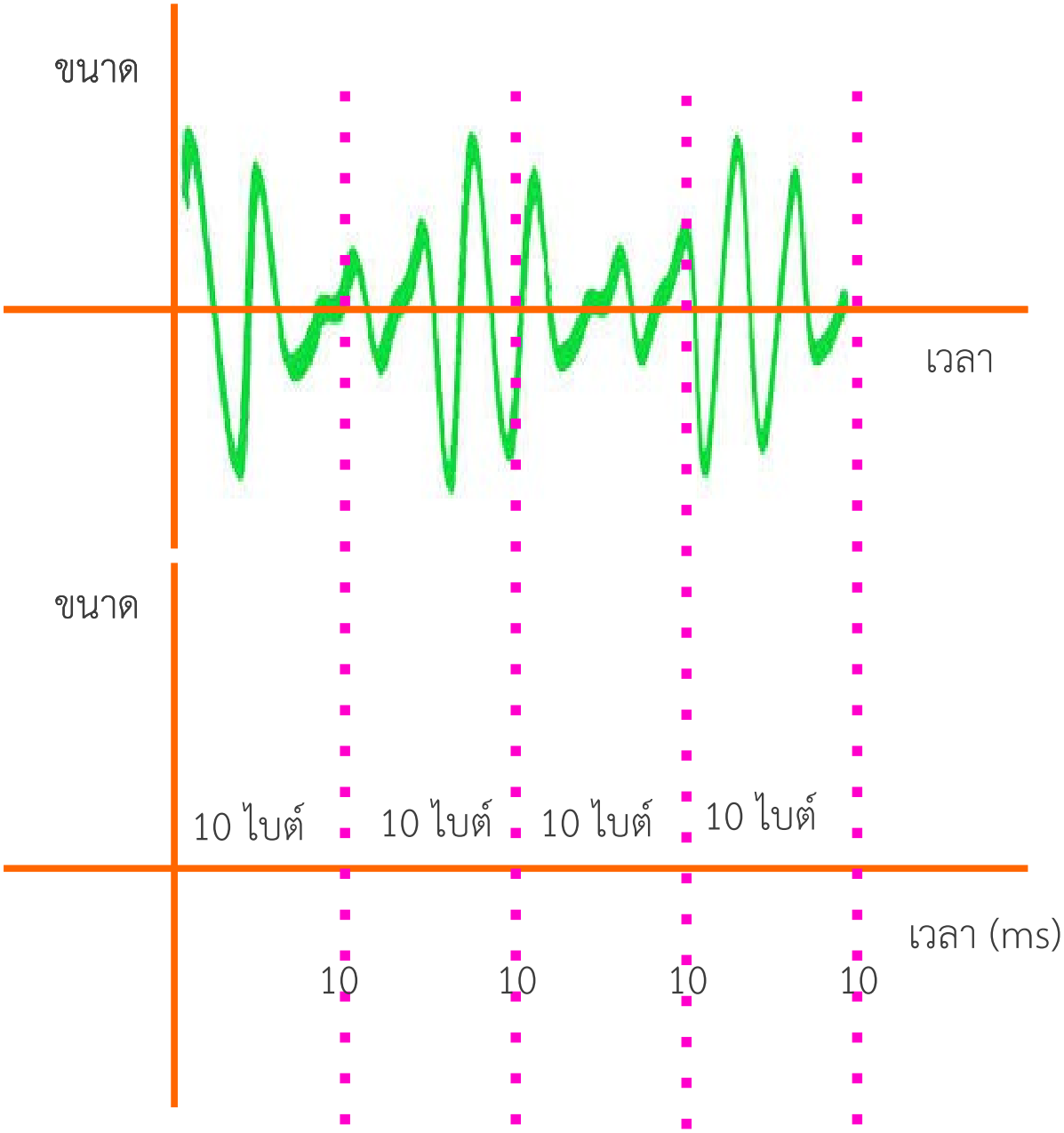
Source Coding(3)



มาตรฐาน G.711

Data rate =?

Source Coding(4)



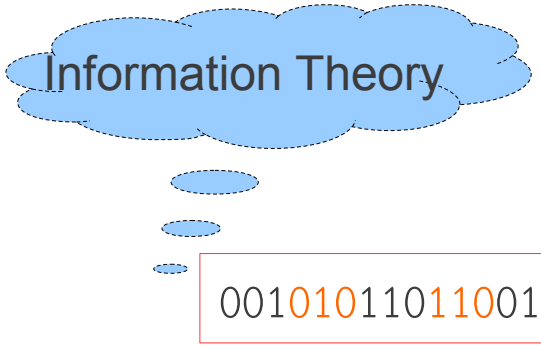
มาตรฐาน G.729

Data rate =?

Compression(5)

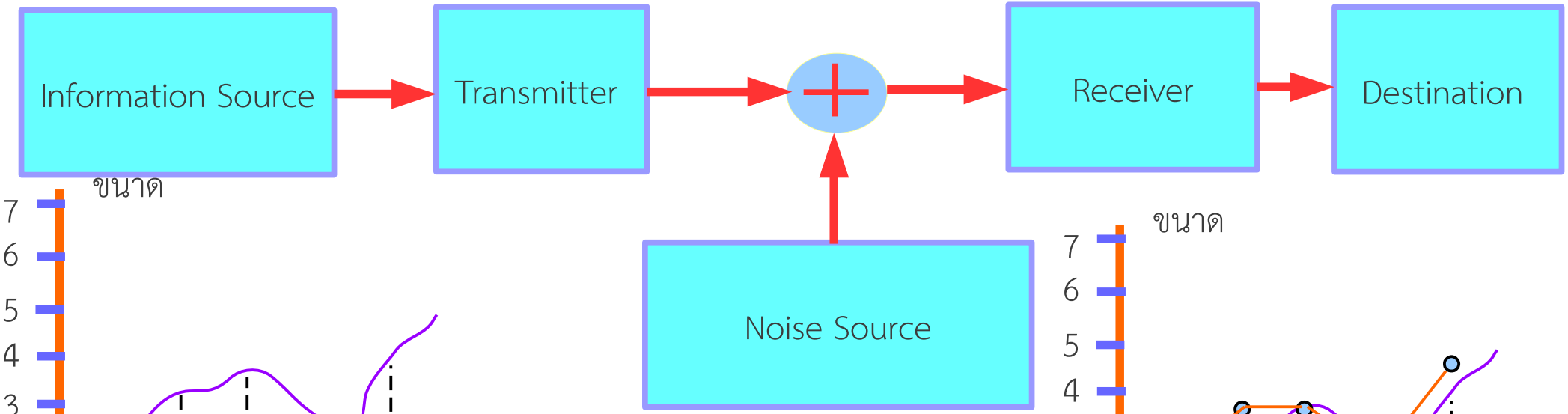
- Lossless Coding
  - เหมือนเดิมเมื่อทำการถอดรหัสกลับ (ถ้าไม่มีการสูญเสียระหว่างสื่อสาร)
- Lossy Coding
  - ไม่เหมือนเดิมเมื่อทำการถอดรหัสกลับ(ถ้าไม่มีการสูญเสียระหว่างสื่อสาร)

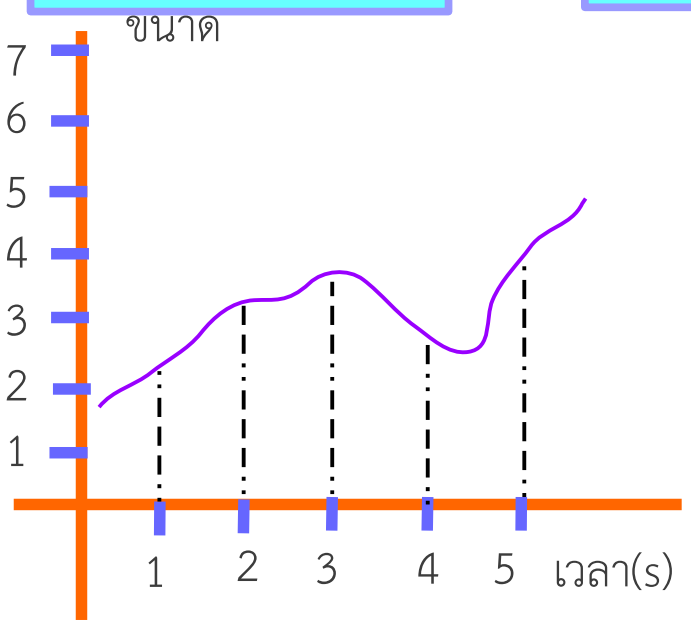
ตัวอย่างการส่งข้อมูล



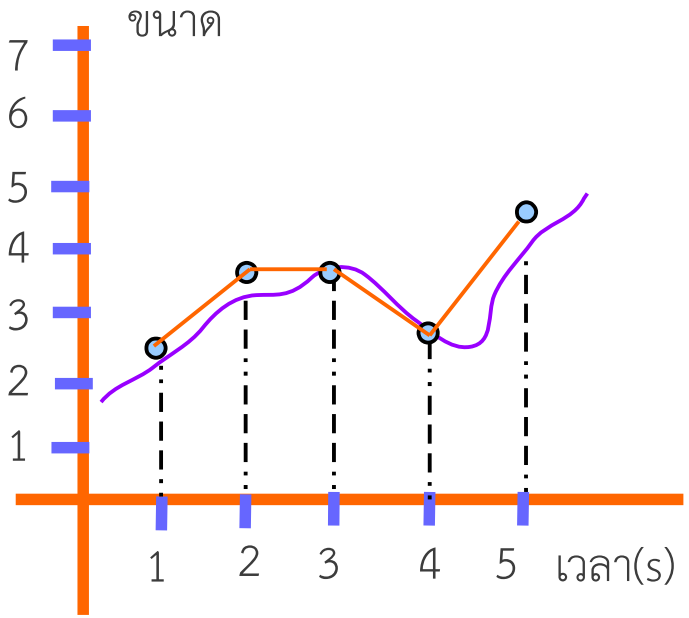
Payload	Header
001010110110010	1....110

001010110110010

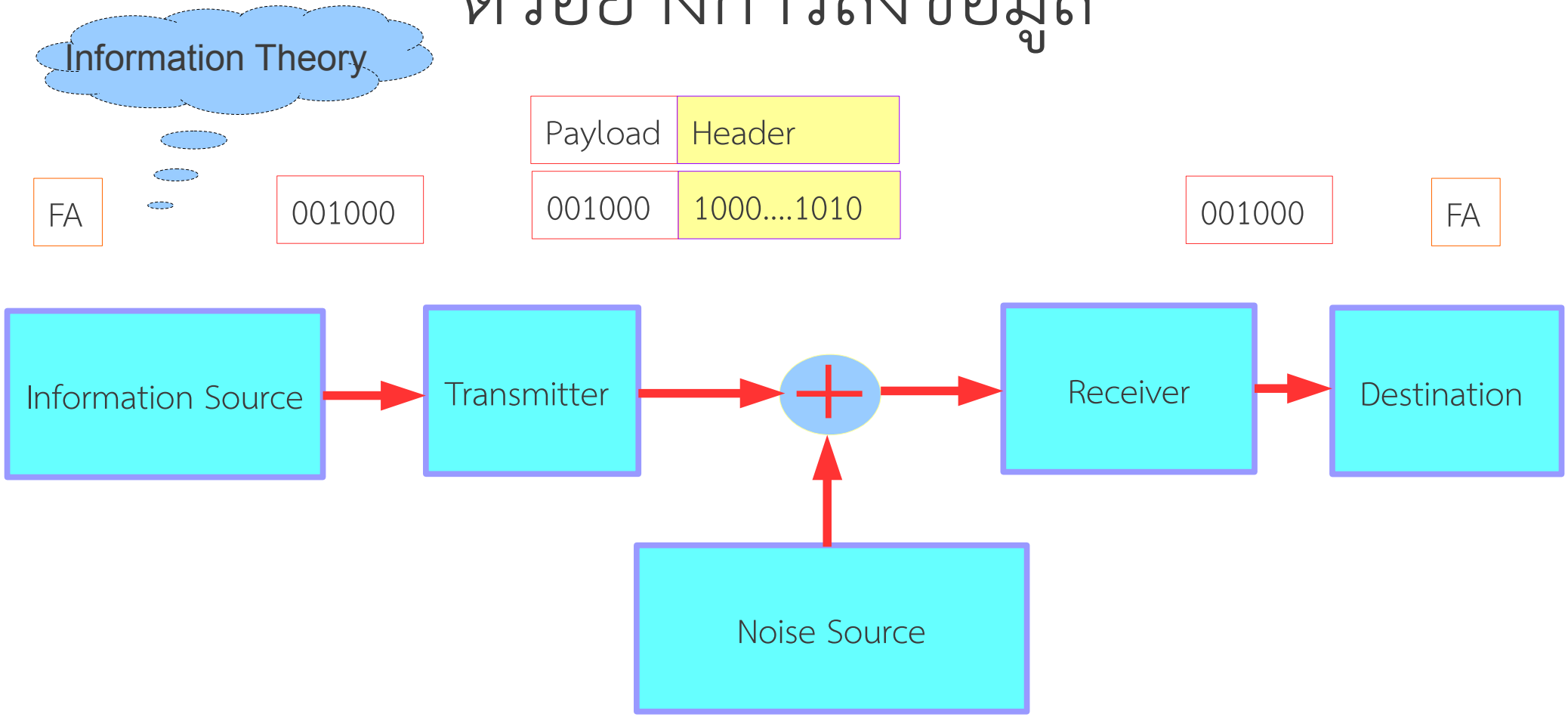




- Lossless Coding or Lossy Coding ?



ตัวอย่างการส่งข้อมูล



- Lossless Coding or Lossy Coding ?

Compression(6)

- Text, Facsimile → Modem
- Audio or Speech, 64 kbps for PSTN, 13 kbps for GSM, etc..
- Video, MPEG4

สรุป

- Information Source
- Entropy
- Source Coding
- Compression
- Next Class: Examples of Information Theory: Source Coding

การบ้าน

- จงอธิบาย
  - การเข้ารหัสเสียง
  - ติดตั้ง SJphone ลงบนเครื่องคอมพิวเตอร์ 2 เครื่องแล้วดักจับเฟรมที่สื่อสารเพื่อใช้ในการโทรหากัน โดยใช้มาตรฐาน SIP ซึ่งเฟรมจะประกอบด้วย
    - เฟรม SIP ใช้ในการสร้างการเชื่อมต่อ
    - เฟรม RTP ใช้สำหรับการส่งข้อมูลเสียง
    - www.sjlabs.com/sjp.html

จงคำนวณต่อไปนี้

Information Content (I)

- Entropy
- Code Efficiency

Source(x)	P(x)	Codeword
A, x_1	1/5	000
B, x_2	3/6	001
C, x_3	1/5	010
D, x_4	1/10	011
F, x_5	0.5/10	100
W, x_6	0.5/10	101